

Mavzu: Eyler integrallari

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1. $\int_0^1 \sqrt{x-x^2}$

2. $\int_0^a \sqrt{a^2-x^2} dx \text{ (a>0)}$

3. $\int (3x+1)^7 dx$

4. $\int \sqrt[3]{1+x^2} x dx$

5. $\int \cos mx dx$

6. $\int (\ln x)^3 \frac{dx}{x}$

7. $\int \frac{\sin \sqrt{x}}{\sqrt{x}} dx$

8. $\int e^{\sin x} \cdot \cos x dx$

9. $\int x \cos x dx$

10. $\int x^2 e^{3x} dx$

11. $\int x^3 \cos 2x dx$

12. $\int_0^\infty \frac{x^2 dx}{1+x^4}$

13. $\int_0^\infty \frac{\sin^2 x}{x^2} dx$

14. $\int_0^\infty \frac{dx}{(1+x^2)^2}$

15. $\int_0^\infty \frac{x^{2n-1}}{(1+x^2)^2} dx \quad (0 < n < 1)$